Heirloom

A dead-man’s switch for your crypto. Check in with Claude. If you go silent, your heirs claim on Solana.

MLH Solana Hackathon, November 2026 · Seunghyeon Baek
· devnet prototype

Self-custody has no plan for the day you are gone

- Keys die with their owner. Families often do not know the coins exist or where the keys are kept.
- A paper will cannot sign a transaction.
- Handing keys to a custodian brings back the trust problem self-custody was meant to solve.

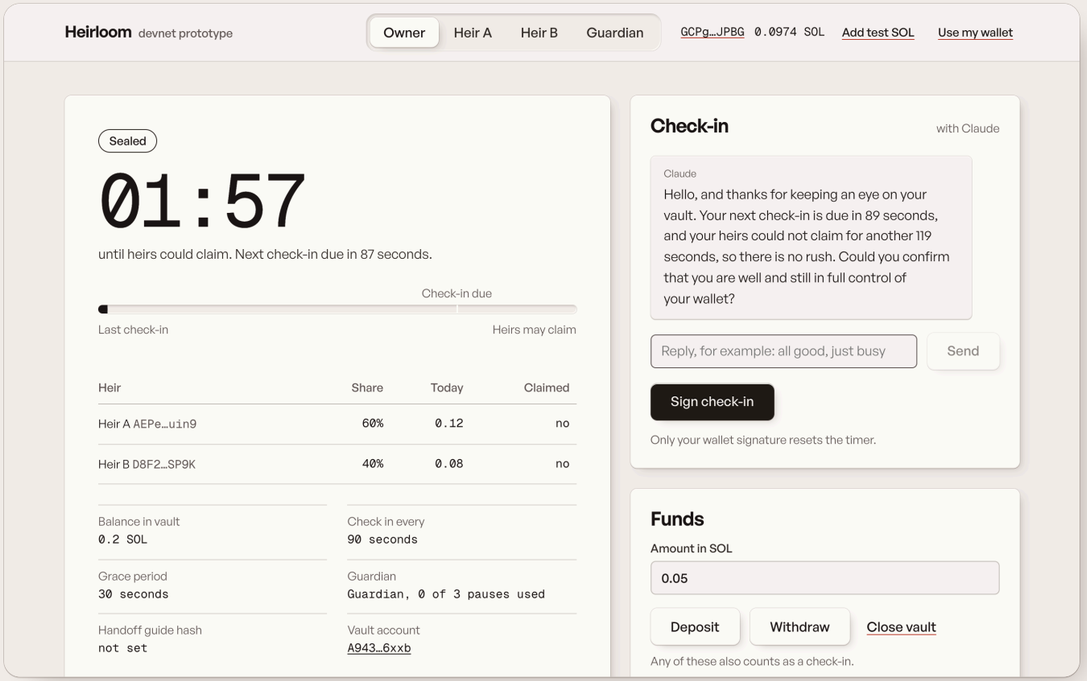
What people need

Full control while alive. A fair, automatic handover when they stop showing up. Plain words for heirs who are not crypto people.

A vault that waits for your check-in

Lock SOL in a program-owned vault. Name up to five heirs and their shares. Choose how often you check in and how long the grace period is.

Miss your check-ins and, after the grace period, each heir can claim their share once. A guardian can delay the unlock but never touch the funds.



Demo flow

1. Create

Owner names Heir A 60% and Heir B 40%, sets a 90 s check-in and 30 s grace, and locks SOL.

`initialize + deposit`

2. Check in

Claude asks how they are. The owner replies and signs. The timer restarts.

`heartbeat`

3. Guide

Rough notes become a handoff guide. Only its SHA-256 goes on-chain.

`set_guide_hash`

4. Silence

No check-in. The countdown passes due, then grace. The vault unlocks.

`Clock sysvar`

5. Claim

Each heir reviews with Claude and claims once. The guide unlocks, verified against the hash.

`claim × 2`

Four demo wallets live in the browser, so anyone can play every role without installing a wallet.

Architecture

```
seeds   = ["vault", owner]
program = 2GvFQtnM7cGa…ZbQbKv
size    = 371 bytes

Vault {
  owner, heartbeat_authority,
  guardian: Option<Pubkey>,
  heirs: [{ key, share_bps }] ≤ 5,
  checkin_interval, grace_period,
  last_heartbeat, paused_until,
  pauses_used, status,
  claimed_bitmap, claim_snapshot,
  guide_hash: [u8; 32], bump
}
lamports = rent minimum + protected SOL
```

Program	Anchor 1.2.1, Rust: 9 instructions, PDA holds lamports
Client	Codama-generated @solana/kit 8 client from the IDL
App	Next.js 16: simulate, then explain, then sign, then confirm
AI	Server route handlers calling Claude (claude-sonnet-5-5); the server reads vault facts from chain
Tests	18 LiteSVM tests + Surfpool end-to-end with time travel

Security model

Who can do what

The owner’s vault PDA is derived from the signer’s key with has_one checks. The relay can only send heartbeats. The guardian can only pause. Heirs can only claim.

Fair claims

The first claim freezes the pot. A bitmap blocks double claims and is set before lamports move. The last heir sweeps the dust. The vault never goes below its rent minimum.

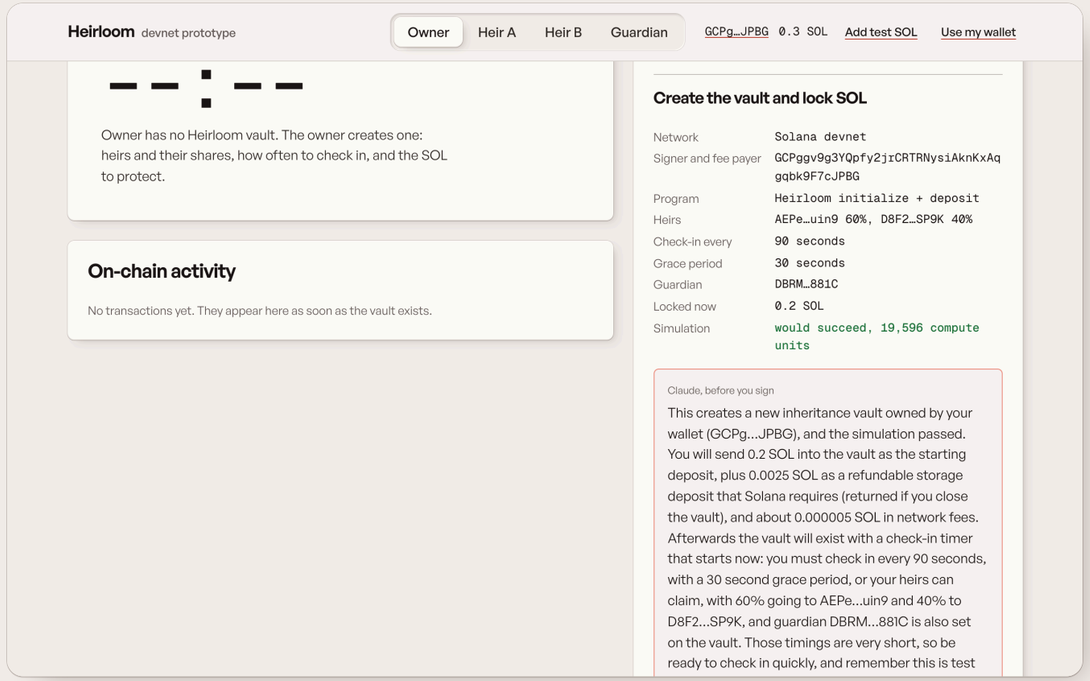
Bounded delay

Each guardian pause lasts at most 30 days, with at most 3 pauses per check-in period. Owner actions revive the vault only before the first claim.

Tested: exact unlock boundary, double claim, wrong signers, a fake vault owned by another program, stray lamports, five heirs in any order.

Claude’s role, with no keys

- **Check-in companion.** It writes the nudge from the vault’s real on-chain phase and responds with care when something is wrong.
- **Explains before you sign.** It reads the simulation and the vault state and says in plain words what moves, to whom, and when to stop.
- **Handoff guide writer.** It turns rough notes into steps for heirs and refuses seed phrases.



Only a wallet signature changes state. The API key stays on the server.

Roadmap and team

- Wallet Standard connection (Phantom, Solflare, Backpack)
- USDC and other SPL tokens held by the vault PDA
- Hosted relay: email and Telegram check-ins, with signed messages triggering a relay heartbeat
- Guides encrypted to heirs, released after a claim
- Multisig guardians and legal wrappers. Heirloom is not a will.

Seunghyeon Baek

Solo builder from South Korea, building in public.

`github.com/seunghyeon1004/heirloom`